



Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

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Motivation

LLMs often fail to produce the diverse, human-like creativity expected in open-ended tasks

=> long-term homogenization of human thought as users are repeatedly exposed to similar outputs

Existing benchmarks are insufficient for evaluating diversity in real-world settings (e.g., random number generation)

INFINITY-CHAT

Real-world open-ended queries with diverse responses

ex) "Write a metaphor about time"

INFINITY-CHAT

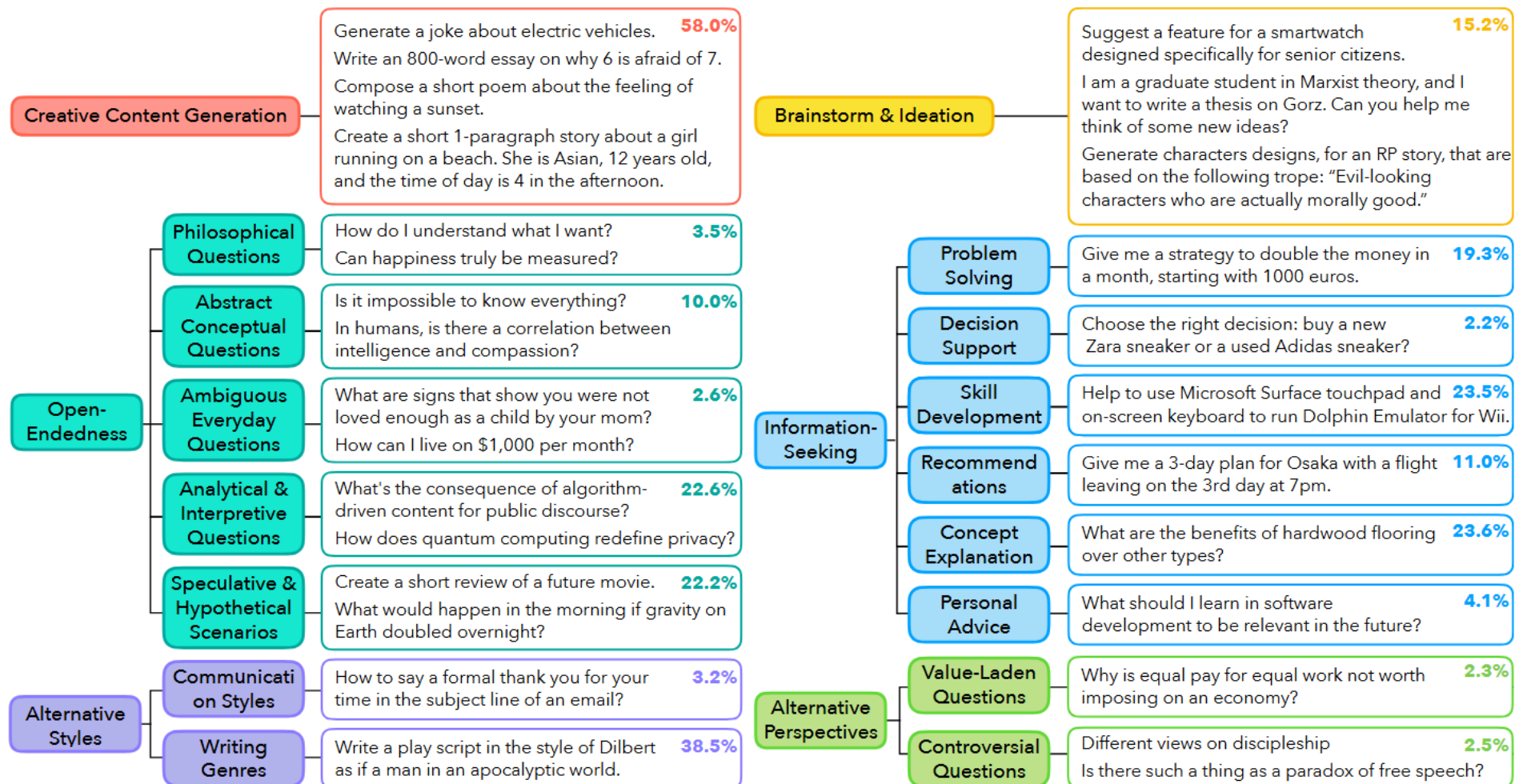
- Source: WildChat
 - single turn GPT-4 queries (English / non-toxic / 15-200 characters)
- Filtering via GPT-4o
 - meaningful information
 - revised ambiguous queries
 - separated into open-ended, closed-ended queries
- 26,070 open-ended queries

Taxonomy of Open-ended queries

Semi-automatic process to construct the taxonomy

- Manually assign tentative labels (~100 queries)
- Iteratively refine and group them into a hierarchical structure
=> 6 high-level categories and 17 fine-grained sub-categories
- Scale the annotation process using GPT-4o

Taxonomy of Open-ended queries



Examine the homogeneity of LLMs

Using a subset of 100 representative queries from INFINITY-CHAT

1. Intra-model repetition: the same LLM fails to generate diverse outputs
2. Inter-model homogeneity: different models produce similar outputs

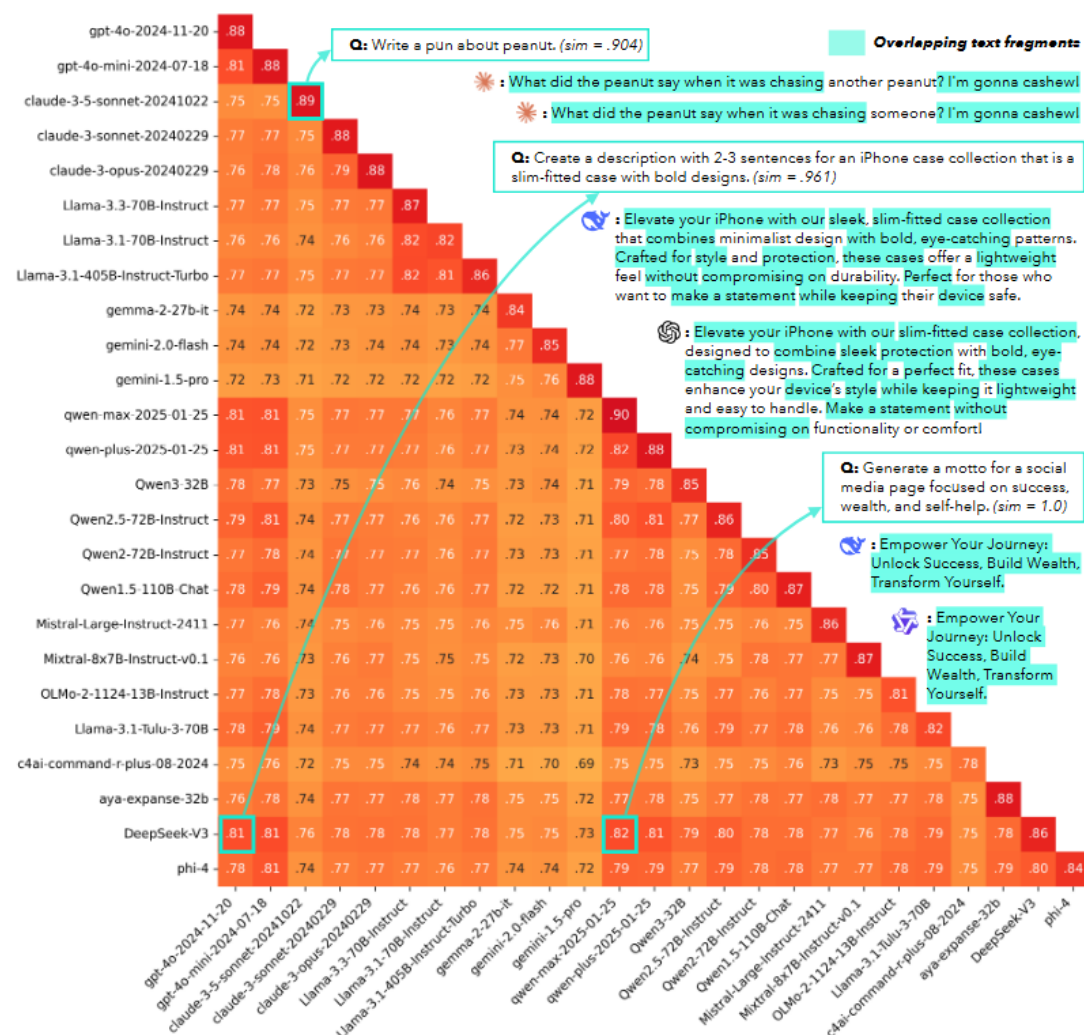
Intra-model Repetition

For each model sample 50 responses per query, compute the average pairwise embeddings similarity within each response pool

	Average	gpt-4o-2024-11-20	gpt-4o-mini-2024-07-18	claude-3.5-sonnet-2024-10-22	claude-3-sonnet-2024-02-29	Llama-3.3-70B-Instruct	Llama-3.1-70B-Instruct	gemma-2-27b-it	gemini-2.0-flash	gemini-1.5-pro	qwen-max-2025-01-25	Qwen-plus-2025-01-25	Qwen3-32B	Qwen2.5-72B-Instruct	Qwen2-72B-Instruct	Qwen1.5-110B-Chat	Mistral-Large-Instruct-2411	Mixtral-8x7B-Instruct-v0.1	OLMo-2-1124-13B-Instruct	Llama-3.1-Tulu-3-70B	c4ai-command-r-plus-08-2024	aya-expense-32b	DeepSeek-V3	phi-4		
0.9-1.0	43.8	51.0	53.0	61.0	48.0	59.0	51.0	23.0	43.0	33.0	40.0	53.0	55.0	56.0	40.0	48.0	35.0	48.0	43.0	45.0	29.0	28.0	24.0	50.0	42.0	38.0
0.8-0.9	35.2	36.0	34.0	22.0	36.0	27.0	30.0	44.0	38.0	43.0	41.0	32.0	37.0	28.0	36.0	26.0	35.0	36.0	34.0	41.0	39.0	39.0	32.0	36.0	39.0	40.0
0.7-0.8	12.6	10.0	9.0	9.0	10.0	7.0	12.0	23.0	11.0	14.0	11.0	11.0	4.0	11.0	14.0	19.0	24.0	10.0	15.0	9.0	12.0	18.0	19.0	8.0	13.0	11.0
0.6-0.7	4.9	1.0	4.0	5.0	5.0	3.0	4.0	5.0	3.0	7.0	4.0	3.0	3.0	3.0	6.0	4.0	5.0	2.0	5.0	4.0	9.0	11.0	10.0	5.0	3.0	9.0
0.5-0.6	1.9	2.0	0.0	2.0	1.0	4.0	1.0	2.0	4.0	0.0	2.0	1.0	1.0	2.0	1.0	1.0	0.0	3.0	2.0	1.0	7.0	1.0	7.0	1.0	0.0	2.0
0.4-0.5	0.7	0.0	0.0	1.0	0.0	0.0	1.0	2.0	0.0	2.0	0.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	2.0	2.0	3.0	0.0	3.0	0.0
0.3-0.4	0.6	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	1.0	2.0	0.0	0.0	0.0	2.0	1.0	1.0	1.0	1.0	0.0	1.0	0.0	3.0	0.0	0.0	0.0
0.2-0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0
0.1-0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0
0.0-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

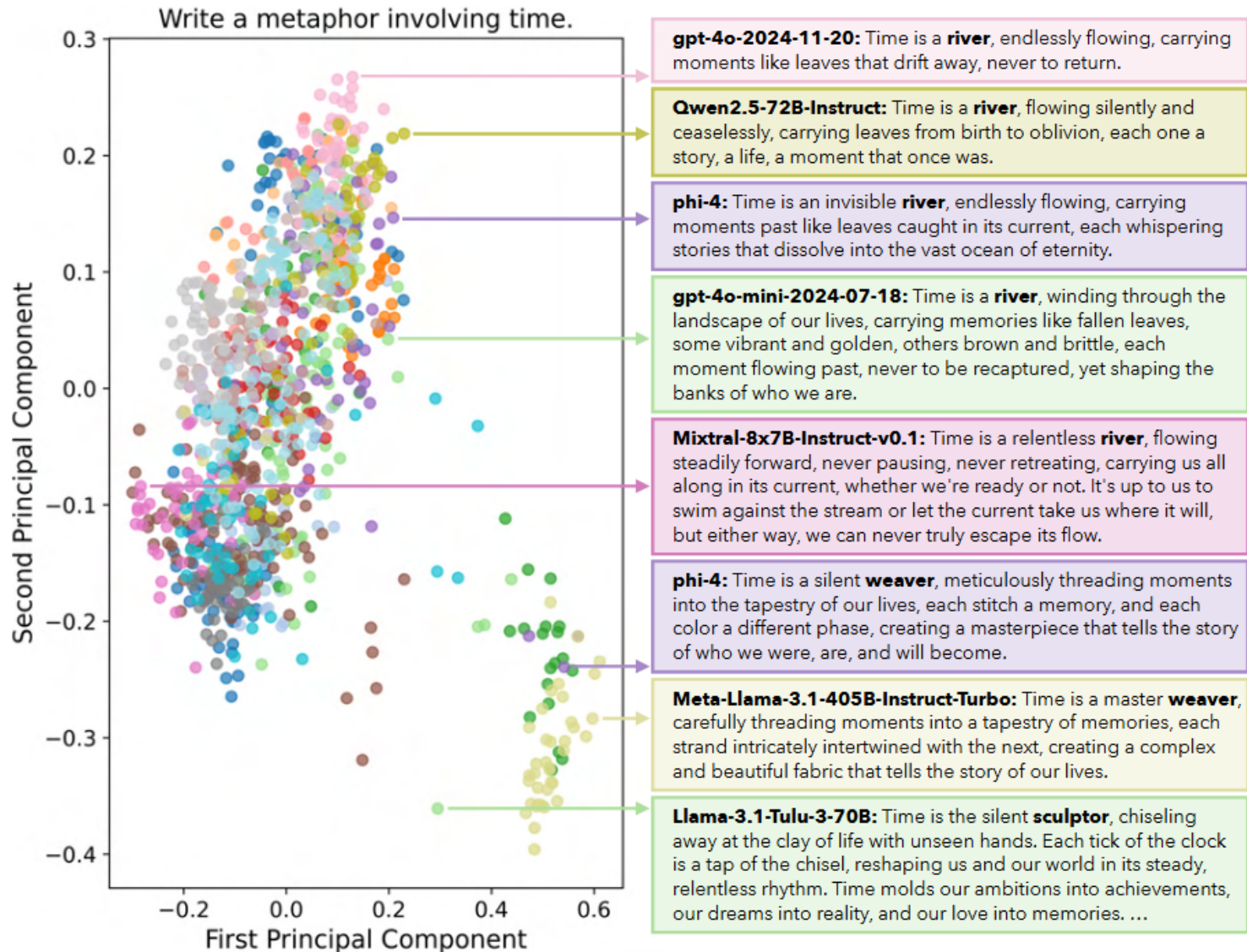
(top-p=0.9, t=1.0) remain highly repetitive

Inter-model Homogeneity



- average pairwise similarity : 71~82%
- GPT and Qwen tend to have higher similarities even with models outside their own families

Syntactic repetition across difference models

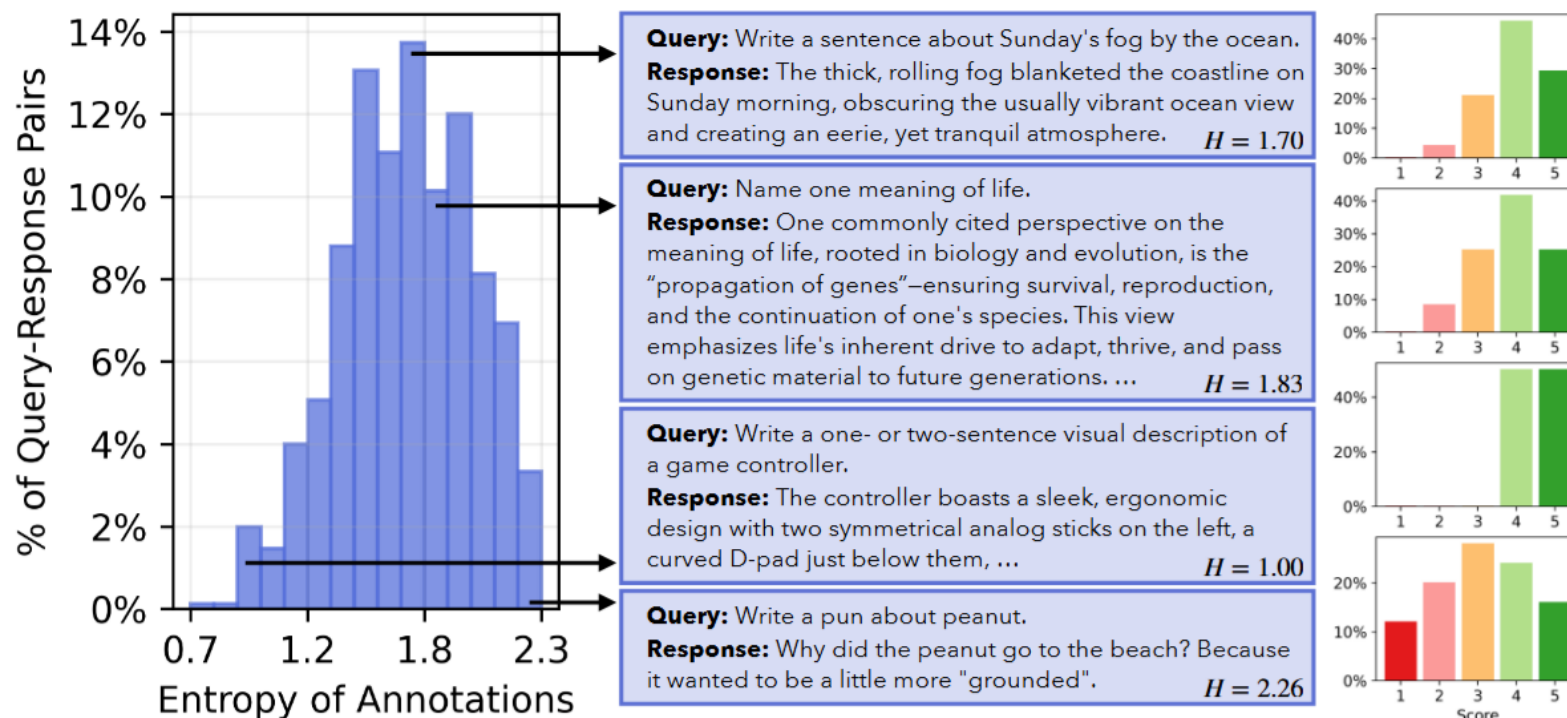


- Surface-level overlap
- Semantically convergence
 - e.g., "river" v.s. "weaver"

How do LLMs handle alternative responses?

Gathering distributional annotations across many humans

1. absolute rating (1-5 scale for response quality): 18,750
2. pairwise preference rating (strong/weak preference between two responses): 12,500



How do LLMs handle alternative responses?

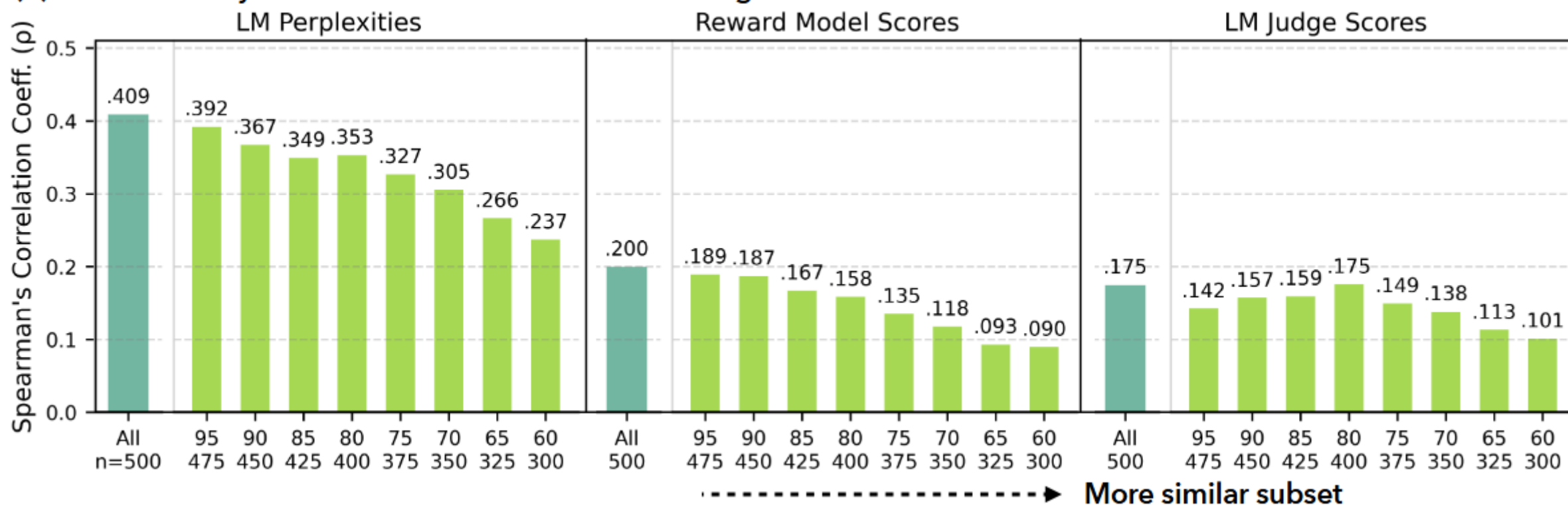
Gathering LLMs, reward models, and judge LLMs ratings

1. ratings of LLMs: derived from response perplexity
2. reward models: standardized scalar reward outputs
3. judge LLMs: prompting (overall, helpfulness, harmlessness, honesty)

Comparing model ratings to human scores

Models show weaker alignment with human ratings for alternative responses of similar quality

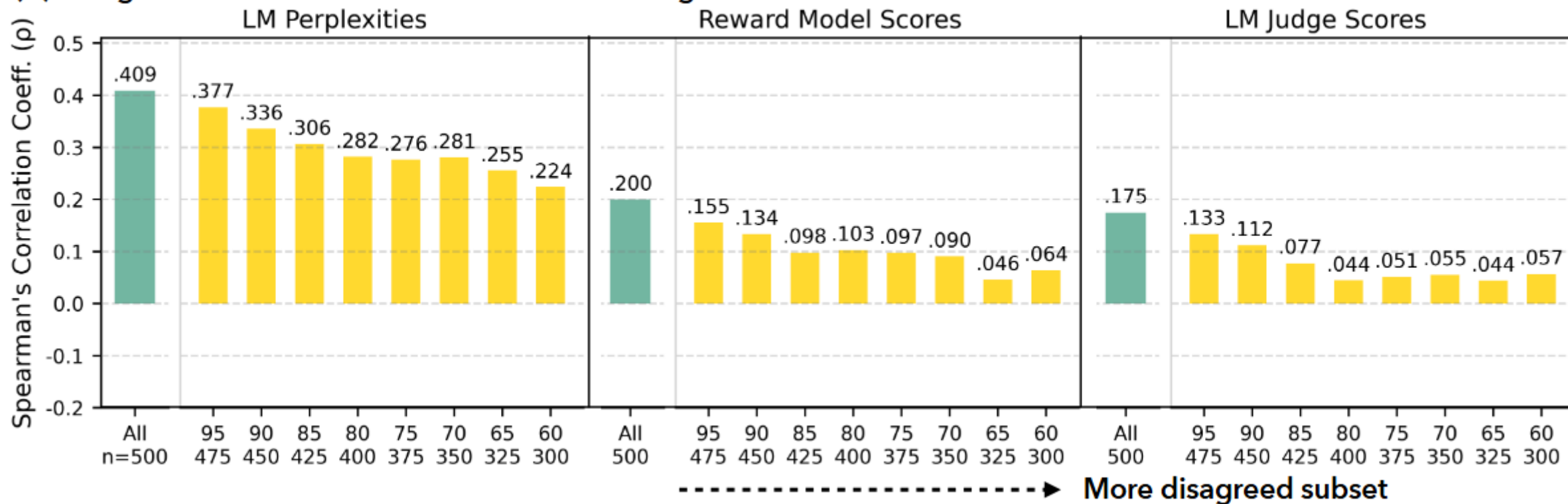
(a) Similar-Quality Subset for Pairwise Preference Rating



Comparing model ratings to human scores

Model judgments are less aligned where annotators disagree

(b) Disagreed Subset for Pairwise Preference Rating



Comparing model ratings to human scores

- Models struggle to equally evaluate similarly high-quality responses at a human level
- Model-human correlation decreases when human annotators disagree more

=> More refined alignment or training methods are needed to reliably assess open-ended responses

Conclusion

- Introduces INFINITY-CHAT
- Comprehensive analysis of the Artificial Hivemind effect
- Diverse taxonomy of prompts with dense human preference annotations